One-meson exchange in the $J^{PC} = 2^{++}$ sector

Partial-wave projection for the $\chi_{b2}(nP)$ extension of Sect. III B and App. D of "Born–Oppenheimer chiral effective field theory and its application to X_b"

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21 October 2026

How to read this note

The paper computes the $J^{PC} = 1^{++}$ system, i.e. the $\chi_{b1}(nP)$ tower coupled to $B^{(*)}\bar{B}^*$. This note does the one-meson-exchange (OME) part of the extension to $J^{PC} = 2^{++}$, the $\chi_{b2}(nP)$ tower: which meson-pair channels exist, which TOPT potentials are needed, the $J = 2$ projectors, and the partial-wave-projected 6×6 OME matrix, in the form of App. D so that it can be copied into the paper. All conventions (charge-conjugation phase, vertices, TOPT denominators, projection formula, normalisations) are the paper's; nothing is redefined. Every statement carries a source tag:

- [P: Eq. (80)] — taken from the paper. Equation, table and section numbers refer to the *current* `Xb.tex`; **the `Xb.pdf` in the project folder is an older build with different numbering.** "l. n" is a line of `Xb.tex`.
- [R: Ref, location] — checked in the cited reference (the arXiv LaTeX source was read).
- [D] — algebra filled in for this note; it follows from the paper's equations. Re-derive it yourself.
- [N] — computed for this note with the scripts in `code/`. **Not in the paper.**

Units: $\hbar = c = 1$, GeV. Energies are measured from the $B\bar{B}^*$ threshold, $E = M - M_{\rm th}$, $M_{\rm th} = m_B + m_{B^*}$ [P: Eq. (4)], also in the 2^{++} sector, so that the TOPT denominators of Eq. (77) can be used unchanged. Potentials are quoted for the exchange of one meson $\phi = \pi, \eta$ and *without* isospin factors, exactly as in Eqs. (D2)–(D10) [P: l. 1051].

Contents

1 The map: what carries over from 1^{++} and what is new **3**

2 The 2^{++} channels **4**

2.1 Which meson pairs are 2^{++} . . . 4

2.2 Thresholds and two-body propagators . . . 4

2.3 Heavy-quark spin and Born–Oppenheimer content . . . 4

3 The OME potentials needed in 2^{++} **5**

3.1 Taken over unchanged . . . 5

3.2 Vanishing transitions . . . 5

3.3 The new transition $B\bar{B} \leftrightarrow B^*\bar{B}^*$ . . . 5

3.4 Summary . . . 6

4 Projectors for $J = 2$ **7**

5 Partial-wave projection **7**

5.1 What a partial-wave projection does 7

5.2 Reduction from two solid angles to one relative angle 8

5.3 From Cartesian contractions to angular moments 10

5.4 A worked example: ${}^5S_2 \to {}^5S_2$ 11

5.5 A worked example with orbital structure: ${}^1D_2 \to {}^5S_2$ 11

5.6 If the remaining angle integral is evaluated analytically 12

5.7 Validation on 1^{++} 13

6 Result: the projected 2^{++} OME **13**

7 Checks **15**

8 The heavy-quark spin symmetry limit **16**

9 What else the 2^{++} LSE needs (outside OME) **18**

10 Porting to `HadronDynamics.jl` **18**

11 Caveats and open points **19**

12 Self-check questions **19**

A Notation and numbers **20**

B Reproduction scripts **20**

C Answers to the self-check questions **21**

1 The map: what carries over from 1^{++} and what is new

At LO the interaction is $V = V^{(\mathrm{OME})} + \sum_{n\le N} V_n^{(\bar{b}b)} + C(\Lambda, N)$ [P: Eq. (10)]. For the $\chi_{b2}(nP)$ system the same three pieces appear; this note supplies the first one in the 2^{++} channel. The paper itself leaves the χ_{b0} and χ_{b2} sectors for the future [P: l. 1748]. The extension is technical because every ingredient already exists: the vertices and TOPT denominators are in the paper, and the $J = 2$ projectors are in Baru *et al.* 2019, App. B, the reference the paper cites for the projection method [P: l. 1083] [R: Baru 2019, App. B]. What has to be done is to collect the channels, supply the one TOPT potential that does not occur in 1^{++}, and carry out the angular algebra.

ingredient	1^{++} (paper)	2^{++} (this note)
Lagrangian, vertices	Eqs. (65)–(70)	unchanged
3-body states, D_β, μ_β	Eqs. (71), (76), (77), $\beta = 1, \ldots, 4$	unchanged
isospin factors	3 (π), $\frac{1}{3}$ (η), Eq. (82)	unchanged
TOPT potentials	$B\bar{B}^* \to B\bar{B}^*$, $B\bar{B}^* \leftrightarrow B^*\bar{B}^*$, $B^*\bar{B}^* \to B^*\bar{B}^*$, Eqs. (72)–(75)	the same four, plus $B\bar{B} \leftrightarrow B^*\bar{B}^*$ (Sect. 3.3); $B\bar{B} \to B\bar{B}$ and $B\bar{B} \leftrightarrow B\bar{B}^*$ vanish
channels	3, Eq. (1)	6, Eq. (1)
projectors	Eq. (D1), one open index	two open indices, Eqs. (6)–(11)
projection	Eq. (80), $1/(2J+1) = 1/3$	Eq. (80), $1/(2J+1) = 1/5$
angular moments $\Delta_m^{(\beta_1\beta_2)}$	Eq. (D11): $m \le 3$; (12), (22), (23), (24)	$m \le 4$; in addition (33), (44)
result	Eqs. (D2)–(D10)	Eqs. (26)–(44), time reversal Eq. (45)

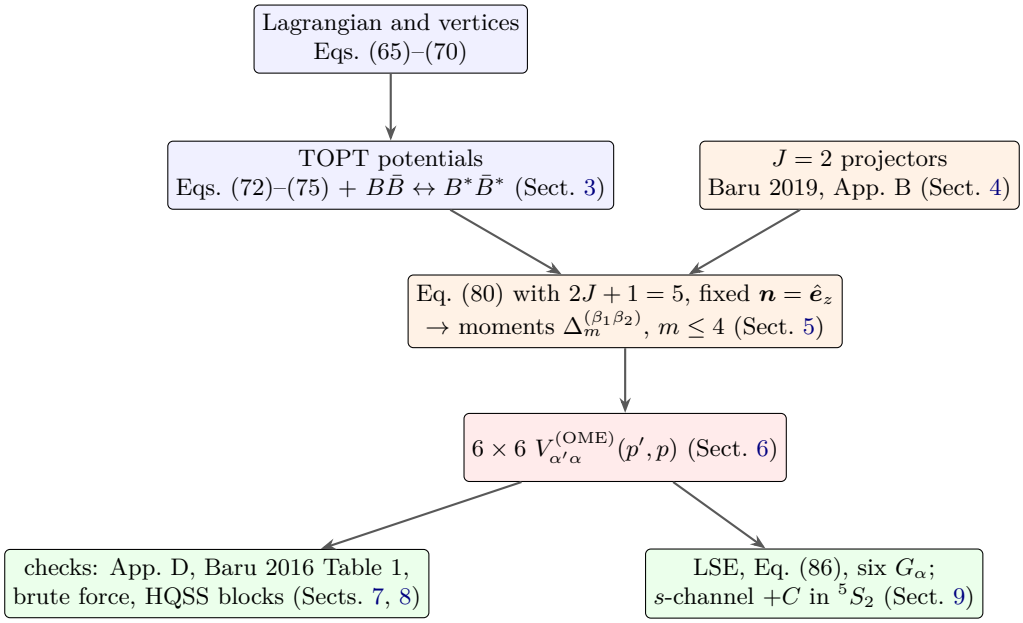


Figure 1: The steps of this note. Blue: input taken over from the paper; orange: the projection; red: the result; green: what it is checked against and where it goes.

Not in this note: the 2^{++} versions of the s-channel exchange, the contact term and the bare χ_{b2} tower. Section 9 lists only what must be changed there to use the OME matrix in the LSE.

2 The 2^{++} channels

2.1 Which meson pairs are 2^{++}

For a meson–antimeson pair with orbital angular momentum L and total spin S: $P = (-1)^L$, so positive parity needs even L; $C = (-1)^{L+S}$ for $B\bar{B}$ and $B^*\bar{B}^*$; and $B\bar{B}^*$ is taken in the C-even combination $(|B\bar{B}^*\rangle - |B^*\bar{B}\rangle)/\sqrt{2}$ of the paper's phase convention $\mathcal{C}V\mathcal{C}^{-1} = -\bar{V}$ [P: I. 952, footnote 14] (derivation of these rules: `notes/s_channel`, Sect. 2.4). Enumerating $J = 2$, $P = +$, $C = +$ [D]:

pair	S	L	$^{2S+1}L_J$	comment
$B\bar{B}$	0	2	$^1D_2$	$S = 0$ forces $L = J = 2$; $C = (-1)^2 = +$
$B\bar{B}^*$, C-even	1	2	$^3D_2$	$L = 0$ gives $J = 1$, $L = 4$ gives $J \geq 3$
$B\bar{B}^*$, C-odd	1	—	—	$C = -$
$B^*\bar{B}^*$	0	2	$^1D_2$	$S = 0$ forces $L = 2$
$B^*\bar{B}^*$	1	—	—	$C = (-1)^{L+1} = +$ needs odd L, $P = +$ needs even L
$B^*\bar{B}^*$	2	0, 2, 4	$^5S_2$, $^5D_2$, $^5G_2$	$\lvert L-2\rvert \leq 2 \leq L+2$
any	≤ 2	≥ 6	—	$J \geq L - S \geq 4$

These six channels exhaust the 2^{++} pairs; they are the basis of Baru *et al.* [R: Baru 2016, Eq. (36); Baru 2017, Eq. (3.11); Baru 2019, Eq. (9)]. They are labelled here

$$\alpha = 1,\dots,6: \quad B^*\bar{B}^*(^5S_2),\ B^*\bar{B}^*(^1D_2),\ B^*\bar{B}^*(^5D_2),\ B^*\bar{B}^*(^5G_2),\ B\bar{B}^*(^3D_2),\ B\bar{B}(^1D_2). \tag{1}$$

Two reasons for this order: (i) $\alpha = 1$ is the S wave, so the projector $\delta_{\alpha' 1}\delta_{\alpha 1}$ of the s-channel and contact terms in Eq. (83) keeps its form; (ii) the four $B^*\bar{B}^*$ channels come first, and their 4×4 block involves a single TOPT combination, $\Delta_m^{(22)}$ (Sect. 6). Baru's order, $B\bar{B}(^1D_2)$, $B\bar{B}^*(^3D_2)$, $B^*\bar{B}^*(^5S_2)$, $B^*\bar{B}^*(^1D_2)$, $B^*\bar{B}^*(^5D_2)$, $B^*\bar{B}^*(^5G_2)$, corresponds to $\alpha = 6, 5, 1, 2, 3, 4$. Below, $^1D_2^*$ means $B^*\bar{B}^*(^1D_2)$ and $^1D_2$ means $B\bar{B}(^1D_2)$ when the two must be distinguished.

2.2 Thresholds and two-body propagators

With $E = M - M_{\rm th}$ and the isospin-limit masses [P: Eqs. (2),(3)] [D]:

α	pair	threshold $- M_{\rm th}$	reduced mass	$G_\alpha(k,q)$
1–4	$B^*\bar{B}^*$	$+\Delta m = +46\,$MeV	$m_{B^*}/2 = 2.663\,$GeV	$\left(q^2/m_{B^*} + \Delta m - E - i\epsilon\right)^{-1}$, the paper's G_3
5	$B\bar{B}^*$	0	$\mu = 2.651\,$GeV	$2\mu/(q^2 - k^2 - i\epsilon)$, the paper's G_1
6	$B\bar{B}$	$-\Delta m = -46\,$MeV	$m_B/2 = 2.640\,$GeV	$\left(q^2/m_B - \Delta m - E - i\epsilon\right)^{-1}$, new

The first two lines are Eq. (87) [P: Eq. (87)]; the third follows in the same way from $2m_B + q^2/m_B - M$. In contrast to 1^{++}, the S-wave channel has the *highest* threshold: channels 5 and 6 are open for $E > 0$ and $E > -\Delta m$. At the $B^*\bar{B}^*$ threshold, $E = \Delta m$, their on-shell momenta are 0.494 and 0.697 GeV [N], of the order of the hard scale; a D-wave suppression of these transitions cannot be expected. (For the charm analogue Baru *et al.* quote ≈ 500 and 700 MeV [R: Baru 2016, Eqs. (40),(41)].) The established $\chi_{b2}(nP)$ lie below all six thresholds: $\chi_{b2}(3P)$ is 10.60 MeV above $\chi_{b1}(3P) = 10.513\,$GeV [P: Table II, I. 1739], i.e. at $\approx 10.524\,$GeV, 34 MeV below $2m_B = 10.558\,$GeV [D].

2.3 Heavy-quark spin and Born–Oppenheimer content

Couple the two heavy spins to $s_{\bar{b}b}$ and the two light spins to j_ℓ. In S wave (`notes/s_channel`, Sect. 2.5, where the table was checked against Bondar *et al.* 2011):

$$|B\bar{B}\rangle = \tfrac{1}{2}\,0_H \otimes 0_\ell + \tfrac{\sqrt{3}}{2}(1_H \otimes 1_\ell)_0, \qquad |B^*\bar{B}^*\rangle_{S=0} = \tfrac{\sqrt{3}}{2}\,0_H \otimes 0_\ell - \tfrac{1}{2}(1_H \otimes 1_\ell)_0,$$
$$|B\bar{B}^*;+\rangle = (1_H \otimes 1_\ell)_1, \qquad |B^*\bar{B}^*\rangle_{S=2} = (1_H \otimes 1_\ell)_2. \tag{2}$$

With even L, $C = C_\ell C_H$ and $P = +$, so a 2^{++} state needs $(s_{\bar{b}b}, j_\ell) = (0,0)$ or $(1,1)$ [P: Eqs. (15),(16)] [D]. Coupling $\boldsymbol{J}_\ell = \boldsymbol{L} + \boldsymbol{j}_\ell$ first [P: l. 493–495]:

$(s_{\bar{b}b}, j_\ell)$	J_ℓ	L	in 1^{++}?
$(0,0)$	2	2	no
$(1,1)$	1	0, 2	yes (the ${}^3S_1$–$(2,1)1$ pair of Eq. (20))
$(1,1)$	2	2	yes ($(2,1)2$ of Eq. (20))
$(1,1)$	3	2, 4	no

Six states, as many as channels [D]. The $J_\ell = 1$ and $J_\ell = 2$ states are common to 1^{++} and 2^{++}; heavy-quark spin symmetry (HQSS) therefore relates the two OME matrices (Sect. 8). New in 2^{++} are the light-spin singlet and $J_\ell = 3$. In Born–Oppenheimer (BO) language the light singlet is Σ_u^-, which cannot contribute to 1^{++} [P: l. 505] but does contribute to 2^{++} with $L = 2$, $s_{\bar{b}b} = 0$; the count is Σ_u^- (1) + $\Sigma_g^{+\prime}$ with $J_\ell = 1, 3$ (2) + Π_g with $J_\ell = 1, 2, 3$ (3) = 6. As in 1^{++}, there is no $\lambda_\ell = 0$ state with $J_\ell = L = 2$ [P: l. 497–504] [D].

3 The OME potentials needed in 2^{++}

3.1 Taken over unchanged

The potentials between C-even $B\bar{B}^*$ and $B^*\bar{B}^*$ states are those of Eqs. (72)–(75) [P: Eqs. (72)–(75)]. They are operators in polarisation space, so they do not know about J; only the projection does. The conventions of l. 1052 carry over: in Eqs. (73) and (74) the two terms are the $B\bar{B}^*$ and $B^*\bar{B}$ components of the C-even state, and $\boldsymbol{A}_a = \frac{i}{\sqrt{2}}(\boldsymbol{\epsilon}_a \times \boldsymbol{\epsilon}_a'^*)$ sits on line $a = 1$ (B^*) or $a = 2$ ($\bar{B}^*$) [P: l. 1052]. The TOPT denominators $D_\beta = 1/\{2E_\phi(q)[E_\phi(q) + \mu_\beta]\}$ and the μ_β are Eqs. (76), (77) [P: Eqs. (76),(77)].

3.2 Vanishing transitions

There is no $BB\phi$ vertex: $0^- \to 0^-0^-$ violates parity, and Eq. (68) has no $P^\dagger P$ or $\bar{P}^\dagger \bar{P}$ term [P: Eq. (68)]. One-meson exchange needs one vertex on each meson line, so

$$V_{B\bar{B}\to B\bar{B}} = V_{B\bar{B}\to B\bar{B}^*} = V_{B\bar{B}^*\to B\bar{B}} = 0 \tag{3}$$

identically, for any masses [D]. Hence $V_{56} = V_{65} = V_{66} = 0$.

3.3 The new transition $B\bar{B} \leftrightarrow B^*\bar{B}^*$

Both lines change $B \to B^*$, through the vertex of Eq. (69). The result, in the paper's conventions [D]:

New TOPT potential (not in the paper)

$$V^{(\mathrm{OME})}_{B\bar{B}\to B^*\bar{B}^*}(\boldsymbol{p}', \boldsymbol{p}; m_\phi) = \frac{g_b^2}{4F_\pi^2}\,(\boldsymbol{\epsilon}_1'^* \cdot \boldsymbol{q})(\boldsymbol{\epsilon}_2'^* \cdot \boldsymbol{q})\,\big[2D_3(\boldsymbol{p}', \boldsymbol{p}; m_\phi)\big], \tag{4}$$

$$V^{(\mathrm{OME})}_{B^*\bar{B}^*\to B\bar{B}}(\boldsymbol{p}', \boldsymbol{p}; m_\phi) = \frac{g_b^2}{4F_\pi^2}\,(\boldsymbol{\epsilon}_1 \cdot \boldsymbol{q})(\boldsymbol{\epsilon}_2 \cdot \boldsymbol{q})\,\big[2D_4(\boldsymbol{p}', \boldsymbol{p}; m_\phi)\big], \tag{5}$$

with $\boldsymbol{q} = \boldsymbol{p} - \boldsymbol{p}'$, $\boldsymbol{\epsilon}_1$ ($\boldsymbol{\epsilon}_2$) the polarisation of the B^* ($\bar{B}^*$), and $D_{3,4}$ of Eqs. (76),(77). The isospin factors are those of Eq. (82).

Derivation [D]

Trace form of the vertices. Write a meson as a 2×2 matrix in heavy⊗light spin, as in $H = P + \boldsymbol{V} \cdot \boldsymbol{\sigma}$ [P: I. 953–958]: $M_B = \frac{1}{\sqrt{2}}$, $M_{B^*}(\boldsymbol{\epsilon}) = \frac{1}{\sqrt{2}} \boldsymbol{\epsilon} \cdot \boldsymbol{\sigma}$ (so $\mathrm{Tr}\, M^\dagger M = 1$), and the same for antimesons. With $\boldsymbol{u} = -\boldsymbol{\nabla}\Phi/F_\pi$, Eq. (65) gives $\frac{g_b}{4F_\pi}\mathrm{Tr}(H^\dagger H \sigma_k)\partial_k\Phi - \frac{g_b}{4F_\pi}\mathrm{Tr}(\sigma_k \bar{H}\bar{H}^\dagger)\partial_k\Phi$. Evaluating the traces reproduces every term of Eq. (68). A meson line therefore contributes $t_1 = \mathrm{Tr}[M_1'^\dagger M_1\, \boldsymbol{\sigma}\cdot\boldsymbol{q}]$ and an antimeson line $t_2 = \mathrm{Tr}[\boldsymbol{\sigma}\cdot\boldsymbol{q}\, \bar{M}_2 \bar{M}_2'^\dagger]$. The overall minus sign of the antimeson term, the factors i from $\partial_k\Phi$, the TOPT sign and the isospin structure are the same for every transition. They combine into one constant K:

$$V_{fi} = K\, t_1\, t_2 \sum\nolimits_{\text{orderings}} D_\beta .$$

The line factors are

	$B \to B$	$B \to B^*$	$B^* \to B$	$B^* \to B^*$
meson, t_1	0	$\boldsymbol{\epsilon}'^* \cdot \boldsymbol{q}$	$\boldsymbol{\epsilon} \cdot \boldsymbol{q}$	$i(\boldsymbol{\epsilon}'^* \times \boldsymbol{\epsilon}) \cdot \boldsymbol{q} = -\sqrt{2}\, \boldsymbol{A} \cdot \boldsymbol{q}$
antimeson, t_2	0	$\boldsymbol{\epsilon}'^* \cdot \boldsymbol{q}$	$\boldsymbol{\epsilon} \cdot \boldsymbol{q}$	$i(\boldsymbol{\epsilon} \times \boldsymbol{\epsilon}'^*) \cdot \boldsymbol{q} = +\sqrt{2}\, \boldsymbol{A} \cdot \boldsymbol{q}$

Calibration on the paper. (a) $B^*\bar{B}^* \to B^*\bar{B}^*$: $t_1 t_2 = -2(\boldsymbol{A}_1 \cdot \boldsymbol{q})(\boldsymbol{A}_2 \cdot \boldsymbol{q})$, with both orderings through $B^*\bar{B}^*\phi$; Eq. (75) fixes $K = g_b^2/(4F_\pi^2)$. (b) $B\bar{B}^* \to B\bar{B}^*$ with $|+\rangle = (|B\bar{B}^*\rangle - |B^*\bar{B}\rangle)/\sqrt{2}$: the direct terms need $B \to B$ or $\bar{B} \to \bar{B}$ on one line and vanish, leaving $-\frac{1}{2}\big[\langle B\bar{B}^*|V|B^*\bar{B}\rangle + \langle B^*\bar{B}|V|B\bar{B}^*\rangle\big] = -K(\boldsymbol{\epsilon}\cdot\boldsymbol{q})(\boldsymbol{\epsilon}'^*\cdot\boldsymbol{q})[D_1 + D_2]$. That is Eq. (72). (c) $B\bar{B}^* \to B^*\bar{B}^*$: $\frac{1}{\sqrt{2}}\big[(\boldsymbol{\epsilon}_1'^* \cdot \boldsymbol{q})(\sqrt{2}\boldsymbol{A}_2 \cdot \boldsymbol{q}) - (-\sqrt{2}\boldsymbol{A}_1 \cdot \boldsymbol{q})(\boldsymbol{\epsilon}_2'^* \cdot \boldsymbol{q})\big] = (\boldsymbol{A}_1 \cdot \boldsymbol{q})(\boldsymbol{\epsilon}_2'^* \cdot \boldsymbol{q}) + (\boldsymbol{A}_2 \cdot \boldsymbol{q})(\boldsymbol{\epsilon}_1'^* \cdot \boldsymbol{q})$, which is Eq. (73). Eq. (74) follows the same way. No free constant or sign is left.

The new element. For $B\bar{B} \to B^*\bar{B}^*$, $t_1 t_2 = (\boldsymbol{\epsilon}_1'^* \cdot \boldsymbol{q})(\boldsymbol{\epsilon}_2'^* \cdot \boldsymbol{q})$. The two time orderings have the intermediate states $B^*(\boldsymbol{p}')\bar{B}(-\boldsymbol{p})\phi$ and $B(\boldsymbol{p})\bar{B}^*(-\boldsymbol{p}')\phi$. Both contain a pseudoscalar with the incoming momentum p and a vector with the outgoing momentum p', which is $\mu_3 = E_B(p) + E_{B^*}(p') - M$ of Eq. (77). Hence $[D_3 + D_3]$ and Eq. (4). Hermitian conjugation, together with $D_3(\boldsymbol{p}, \boldsymbol{p}') = D_4(\boldsymbol{p}', \boldsymbol{p})$ [P: I. 2137], gives Eq. (5).

Comparison with the literature. Baru *et al.* have the same structure, $(\boldsymbol{\epsilon}_1'^* \cdot \boldsymbol{q})(\boldsymbol{\epsilon}_2'^* \cdot \boldsymbol{q})\, 2/D_{BB^*\pi}$, with $D_{BB^*\pi}$ the μ_3 denominator [R: Baru 2017, Eqs. (3.5),(3.7); Baru 2019, Eqs. (43),(47)]. Their normalisation differs: they absorb the integration weight into V, use the isovector factor, and in 2017 write $\boldsymbol{q} = \boldsymbol{p} + \boldsymbol{p}'$. Their relative sign between $V_{B\bar{B}\to B^*\bar{B}^*}$ and $V_{B^*\bar{B}^*\to B^*\bar{B}^*}$ is opposite to ours, because they use $\mathcal{C}\hat{M} = \bar{M}$ for all mesons [R: Baru 2017, Eq. (2.2); Baru 2019, Eqs. (10),(11)], while the paper uses $\mathcal{C}V\mathcal{C}^{-1} = -\bar{V}$ [P: I. 952]. The two conventions differ by the phase of every $\bar{B}^*$ state. That flips $V_{B\bar{B}\to B^*\bar{B}^*}$ and leaves Eqs. (72)–(75) unchanged; it is equivalent to flipping the sign of the $B\bar{B}$ basis state, so no observable changes [D].

3.4 Summary

out ← in	spin structure	prefactor	$[D_{\beta_1} + D_{\beta_2}]$	
$B\bar{B}^* \leftarrow B\bar{B}^*$	$(\boldsymbol{\epsilon}\cdot\boldsymbol{q})(\boldsymbol{\epsilon}'^*\cdot\boldsymbol{q})$	$-g_b^2/4F_\pi^2$	$D_1 + D_2$	Eq. (72)
$B^*\bar{B}^* \leftarrow B\bar{B}^*$	$(\boldsymbol{A}_1\cdot\boldsymbol{q})(\boldsymbol{\epsilon}_2'^*\cdot\boldsymbol{q}) + (\boldsymbol{A}_2\cdot\boldsymbol{q})(\boldsymbol{\epsilon}_1'^*\cdot\boldsymbol{q})$	$+g_b^2/4F_\pi^2$	$D_3 + D_2$	Eq. (73)
$B\bar{B}^* \leftarrow B^*\bar{B}^*$	$(\boldsymbol{A}_1\cdot\boldsymbol{q})(\boldsymbol{\epsilon}_2\cdot\boldsymbol{q}) + (\boldsymbol{A}_2\cdot\boldsymbol{q})(\boldsymbol{\epsilon}_1\cdot\boldsymbol{q})$	$+g_b^2/4F_\pi^2$	$D_4 + D_2$	Eq. (74)
$B^*\bar{B}^* \leftarrow B^*\bar{B}^*$	$(\boldsymbol{A}_1\cdot\boldsymbol{q})(\boldsymbol{A}_2\cdot\boldsymbol{q})$	$-g_b^2/2F_\pi^2$	$2D_2$	Eq. (75)
$B^*\bar{B}^* \leftarrow B\bar{B}$	$(\boldsymbol{\epsilon}_1'^*\cdot\boldsymbol{q})(\boldsymbol{\epsilon}_2'^*\cdot\boldsymbol{q})$	$+g_b^2/4F_\pi^2$	$2D_3$	Eq. (4)
$B\bar{B} \leftarrow B^*\bar{B}^*$	$(\boldsymbol{\epsilon}_1\cdot\boldsymbol{q})(\boldsymbol{\epsilon}_2\cdot\boldsymbol{q})$	$+g_b^2/4F_\pi^2$	$2D_4$	Eq. (5)
$B\bar{B} \leftarrow B\bar{B}$, $B\bar{B} \leftrightarrow B\bar{B}^*$	—	0	—	

4 Projectors for $J = 2$

Baru *et al.* 2019 list a complete set of projectors for 0^{++}, 1^{+-}, 1^{++} and 2^{++} [R: Baru 2019, App. B, Eqs. (B1)–(B19)]. Their 1^{++} projectors (B5), (B6), (B12) coincide with the paper's Eq. (D1), signs included [R: Baru 2019, Eqs. (B5),(B6),(B12)] [P: Eq. (D1)]. Their 2^{++} projectors therefore extend the paper's phase conventions, and are used here unchanged. With $\boldsymbol{n} = \hat{\boldsymbol{p}}$,

$$P_{ij}\big(B^*\bar{B}^*({}^5S_2)\big) = \tfrac{1}{2}\, S_{ij}, \tag{6}$$

$$P_{ij}\big(B^*\bar{B}^*({}^1D_2)\big) = -\sqrt{\tfrac{5}{2}}\,(\boldsymbol{\epsilon}_1\cdot\boldsymbol{\epsilon}_2)\, v_{ij}, \tag{7}$$

$$P_{ij}\big(B^*\bar{B}^*({}^5D_2)\big) = -\sqrt{\tfrac{45}{56}}\Big(S_{ik}v_{kj} + S_{jk}v_{ki} - \tfrac{2}{3}\delta_{ij}S_{kl}v_{kl}\Big), \tag{8}$$

$$P_{ij}\big(B^*\bar{B}^*({}^5G_2)\big) = \sqrt{\tfrac{175}{32}}\, S_{kl}\, v_{ijkl}, \tag{9}$$

$$P_{ij}\big(B\bar{B}^*({}^3D_2)\big) = -\tfrac{\sqrt{5}}{2}\,\epsilon_k\big(i\varepsilon_{ikl}v_{lj} + i\varepsilon_{jkl}v_{li}\big), \tag{10}$$

$$P_{ij}\big(B\bar{B}({}^1D_2)\big) = -\sqrt{\tfrac{15}{2}}\, v_{ij}, \tag{11}$$

[R: Baru 2019, Eqs. (B10),(B11),(B13),(B14),(B7),(B4)], where [R: Baru 2019, Eqs. (B15)–(B18)]

$$\begin{aligned} v_{ij} &= n_in_j - \tfrac{1}{3}\delta_{ij}, \qquad S_{ij} = \epsilon_{1i}\epsilon_{2j} + \epsilon_{1j}\epsilon_{2i} - \tfrac{2}{3}\delta_{ij}\,\boldsymbol{\epsilon}_1\cdot\boldsymbol{\epsilon}_2, \\ v_{ijkl} &= n_in_jn_kn_l - \tfrac{1}{7}\big(n_in_j\delta_{kl} + n_in_k\delta_{jl} + n_in_l\delta_{jk} + n_jn_k\delta_{il} + n_jn_l\delta_{ik} + n_kn_l\delta_{ij}\big) \\ &\qquad + \tfrac{1}{35}\big(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\big). \end{aligned} \tag{12}$$

How to read them (cf. `notes/s_channel`, Sect. 2.9):

- the open pair ij is symmetric and traceless: five components, the polarisation tensor of the $J = 2$ state. This is why $J = 2$ projectors have two open indices;
- v_{ij} and v_{ijkl} are the rank-2 and rank-4 harmonic tensors of $\boldsymbol{n}$, i.e. $L = 2$ and $L = 4$; no $\boldsymbol{n}$ means $L = 0$;
- $\boldsymbol{\epsilon}_1\cdot\boldsymbol{\epsilon}_2$ is the $S = 0$ state of two vector mesons, S_{ij} the $S = 2$ state, and a single $\boldsymbol{\epsilon}$ is $S = 1$;
- $i\varepsilon_{ikl}\epsilon_k v_{lj}$ couples $S = 1$ and $L = 2$ to a rank-2 tensor, symmetrised in ij and automatically traceless because $\varepsilon_{ikl}v_{li} = 0$; $S_{ik}v_{kj} + \ldots$ is the $J = 2$ part of $S = 2 \otimes L = 2$; $S_{kl}v_{ijkl}$ contracts two of the four $L = 4$ indices with $S = 2$.

Each tensor has definite L, S and $J = 2$. Since each (L, S, J) occurs once, the projectors are the $|(LS)J\rangle$ states up to phases. With $1/(2J + 1) = 1/5$ in Eq. (81), all six are normalised, and the four $B^*\bar{B}^*$ projectors are mutually orthogonal [N]. The phases matter only for comparisons with standard recoupling tables; see the box at the end of Sect. 8.

5 Partial-wave projection

5.1 What a partial-wave projection does

The unprojected potential is a matrix element between states with specified *directions* and meson spins, $V^{(\mathrm{OME})}_{\sigma'\sigma}(p'\boldsymbol{n}', p\boldsymbol{n})$, where σ denotes all incoming spin labels and σ' all outgoing ones. A partial-wave state instead has specified L, S, J and magnetic quantum number M_J. It is a superposition of momentum directions and spin states, not a state with one special direction. The projection is simply the matrix element of the same operator in this new basis. The magnitudes p, p' are not integrated out, and need not be equal.

To see the basis change explicitly, use spin–angular wave functions normalised with the paper's measure $d\Omega/(4\pi)$:

$$\Phi^J_{\alpha M_J}(\boldsymbol{n}, \sigma) = \sqrt{4\pi} \sum_{m_L, m_S} \langle Lm_L, Sm_S | JM_J\rangle\; Y_{Lm_L}(\boldsymbol{n})\, \chi_{Sm_S}(\sigma), \tag{13}$$

where α specifies the meson species and (L, S). The χ_{Sm_S} are normalised coupled-spin wave functions. The factor $\sqrt{4\pi}$ compensates for using $d\Omega/(4\pi)$ rather than $d\Omega$; with the standard spherical harmonics,

$$\sum_{\sigma} \int \frac{d\Omega_{\boldsymbol{n}}}{4\pi}\, \Phi^{J*}_{\alpha' M'_J}(\boldsymbol{n}, \sigma) \Phi^{J}_{\alpha M_J}(\boldsymbol{n}, \sigma) = \delta_{\alpha'\alpha}\delta_{M'_J M_J}.$$

Channel phases are chosen to agree with the projectors in Sect. 4. The partial-wave matrix element is therefore [D]

$$\begin{aligned} \sum_{\sigma',\sigma} \int \frac{d\Omega_{\boldsymbol{n}'}}{4\pi} \frac{d\Omega_{\boldsymbol{n}}}{4\pi}\, \Phi^{J*}_{\alpha' M'_J}(\boldsymbol{n}', \sigma')\, V^{(\mathrm{OME})}_{\sigma'\sigma}(p'\boldsymbol{n}', p\boldsymbol{n})\, \Phi^{J}_{\alpha M_J}(\boldsymbol{n}, \sigma) \\ = \delta_{M'_J M_J}\, V^{(\mathrm{OME})}_{\alpha'\alpha}(p', p). \end{aligned} \tag{14}$$

Rotational invariance makes this diagonal in M_J and independent of its value. It does *not* force $\alpha' = \alpha$: different (L, S) with the same J can mix.

Why use the Cartesian projectors? The tensors P_{ij} of Sect. 4 encode these same $J = 2$ spin–angular wave functions without writing Clebsch–Gordan sums. To understand the two open indices, introduce five orthonormal symmetric-traceless polarisation tensors $e^{(M_J)}_{ij}$, $M_J = -2, \ldots, 2$. Their completeness relation is

$$\sum_{M_J=-2}^{2} e^{(M_J)}_{ij} e^{(M_J)*}_{kl} = \Pi_{ij,kl}, \qquad \Pi_{ij,kl} = \tfrac{1}{2}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) - \tfrac{1}{3}\delta_{ij}\delta_{kl}. \tag{15}$$

Explicitly, in this basis the angular wave functions can be written as

$$\Phi^{2}_{\alpha M_J}(\boldsymbol{n}, \sigma) = \sum_{i,j} P_{ij}(\alpha; \boldsymbol{n}, \sigma)\, e^{(M_J)}_{ij}.$$

Since every P_{ij} is symmetric and traceless, $\Pi_{ij,kl} P_{kl} = P_{ij}$. Consequently, summing over the five M_J states is equivalent to contracting the Cartesian indices ij. After also contracting the meson-spin indices, this is precisely

$$\mathrm{Tr}[P^{\dagger} V^{(\mathrm{OME})} P] = \sum_{i,j} \sum_{\sigma',\sigma} P^{*}_{ij}(\alpha'; \boldsymbol{n}', \sigma')\, V^{(\mathrm{OME})}_{\sigma'\sigma}(p'\boldsymbol{n}', p\boldsymbol{n})\, P_{ij}(\alpha; \boldsymbol{n}, \sigma). \tag{16}$$

Here $P_{ij}(\alpha; \boldsymbol{n}, \sigma)$ means the components in a Cartesian meson-spin basis; equivalently one can use polarisation vectors and the completeness sums $\sum_{\lambda} \epsilon_a(\lambda)\epsilon^{*}_b(\lambda) = \delta_{ab}$. The word "projector" denotes the spin–angular tensors here; no multiplication $P^2 = P$ is needed.

Averaging the identical diagonal matrix elements of Eq. (14) over M_J gives the paper's formula, Eq. (80), with $J = 2$ [P: Eqs. (80),(81)]:

$$V^{(\mathrm{OME})}_{\alpha'\alpha}(p', p; m_\phi) = \frac{1}{5} \int \frac{d\Omega_{\boldsymbol{n}'}}{4\pi} \frac{d\Omega_{\boldsymbol{n}}}{4\pi}\, \mathrm{Tr}\Big[P^{\dagger}(\alpha'; \boldsymbol{n}')\, V^{(\mathrm{OME})}(p'\boldsymbol{n}', p\boldsymbol{n}; m_\phi)\, P(\alpha; \boldsymbol{n})\Big]. \tag{17}$$

The factor $1/5$ removes the sum over the *five total-J states*. It is not an average over nine independent ij components, nor an average over all nine spin states of two vector mesons: the projectors have already selected the requested spin–orbital channel.

5.2 Reduction from two solid angles to one relative angle

The important distinction is between the *overall orientation* of the pair $(\boldsymbol{n}, \boldsymbol{n}')$ and their *relative angle*. Define the fully contracted integrand, including the $1/5$, by

$$\mathcal{F}_{\alpha'\alpha}(\boldsymbol{n}', \boldsymbol{n}) = \frac{1}{5}\mathrm{Tr}[P^{\dagger}(\alpha'; \boldsymbol{n}')\, V^{(\mathrm{OME})}(p'\boldsymbol{n}', p\boldsymbol{n})\, P(\alpha; \boldsymbol{n})].$$

The fixed parameters p, p', m_ϕ are suppressed in this paragraph. Under a simultaneous rotation R of both momentum directions and the spin tensors, all contracted rotation matrices cancel. Thus [D]

$$\mathcal{F}_{\alpha'\alpha}(R\boldsymbol{n}', R\boldsymbol{n}) = \mathcal{F}_{\alpha'\alpha}(\boldsymbol{n}', \boldsymbol{n}) = f_{\alpha'\alpha}(x), \qquad x = \boldsymbol{n}\cdot\boldsymbol{n}' = \cos\theta. \tag{18}$$

There is no preferred spatial direction left. For fixed magnitudes, a pair of unit vectors has only one rotational invariant, x. **This statement applies after the complete contraction in Eq. (16), not to each projector separately.**

Step 1: fix the incoming direction. For any given $\boldsymbol{n}$, choose a rotation $R_{\boldsymbol{n}}$ with $R_{\boldsymbol{n}}\boldsymbol{n} = \hat{\boldsymbol{e}}_z$, and change the outgoing integration variable to $\boldsymbol{m} = R_{\boldsymbol{n}}\boldsymbol{n}'$. Rotations preserve solid angle, so

$$\int \frac{d\Omega_{\boldsymbol{n}'}}{4\pi}\,\mathcal{F}(\boldsymbol{n}', \boldsymbol{n}) = \int \frac{d\Omega_{\boldsymbol{m}}}{4\pi}\,\mathcal{F}(\boldsymbol{m}, \hat{\boldsymbol{e}}_z).$$

The right-hand side is independent of the original $\boldsymbol{n}$. Its remaining average over $\boldsymbol{n}$ therefore multiplies it by $\int d\Omega_{\boldsymbol{n}}/(4\pi) = 1$. This proves, rather than assumes, that the double average can be evaluated at $\boldsymbol{n} = \hat{\boldsymbol{e}}_z$.

Step 2: remove the outgoing azimuth. In this coordinate system,

$$\boldsymbol{n}' = (\sqrt{1-x^2}\cos\varphi,\ \sqrt{1-x^2}\sin\varphi,\ x), \qquad \frac{d\Omega_{\boldsymbol{n}'}}{4\pi} = \frac{d\varphi}{2\pi}\frac{dx}{2}.$$

Here the outgoing polar angle is exactly the angle θ *between* $\boldsymbol{p}$ and $\boldsymbol{p}'$. Since the scalar $f(x)$ has no azimuthal dependence, $\int_0^{2\pi} d\varphi/(2\pi) = 1$, leaving

$$\boxed{\int \frac{d\Omega_{\boldsymbol{n}}}{4\pi}\frac{d\Omega_{\boldsymbol{n}'}}{4\pi}\,\mathcal{F}(\boldsymbol{n}', \boldsymbol{n}) = \frac{1}{2}\int_{-1}^{1} dx\, f(x).} \tag{19}$$

The $1/2$ is the normalisation of the remaining angular measure; it is unrelated to the $1/5$ spin–angular trace. For example, a constant integrand gives $\frac{1}{2}\int_{-1}^{1} dx\, 1 = 1$, as must the original two normalised sphere averages.

What has, and has not, been fixed

Fixing $\boldsymbol{n} = \hat{\boldsymbol{e}}_z$ does *not* set $L = 0$: a D-wave projector is still evaluated at that direction and contributes its nontrivial tensor. Nor may one set both $\boldsymbol{n}$ and $\boldsymbol{n}'$ to $\hat{\boldsymbol{e}}_z$, which would keep only $x = 1$. Only the arbitrary orientation of the whole momentum pair has been removed; all relative angles remain in the integral. A single uncontracted tensor component, or the integrand at a fixed M_J before the sum, need not be a rotational scalar and does not justify this shortcut.

Connection to the familiar Legendre projection. For spinless particles, $J = L$, and the spherical-harmonic addition theorem gives

$$\frac{1}{2L+1}\sum_{M_J=-L}^{L}\left[\sqrt{4\pi}Y_{LM_J}(\boldsymbol{n}')\right]^*\left[\sqrt{4\pi}Y_{LM_J}(\boldsymbol{n})\right] = P_L(x).$$

For a central spinless kernel the same argument therefore gives $V_L(p', p) = \frac{1}{2}\int_{-1}^{1} dx\, P_L(x)V(p', p, x)$ in the present normalisation. In our spinful coupled-channel problem, the contraction $P^\dagger V^{(\mathrm{OME})}P$ supplies the spin–angular weight instead. One must *not* multiply every $J = 2$ matrix element by an additional $P_2(x)$: that would project the angular dependence twice.

5.3 From Cartesian contractions to angular moments

Now separate a TOPT potential into the scalar prefactor c of Sect. 3.4, its spin–momentum operator $\mathcal{O}(\boldsymbol{q})$, and its denominator combination:

$$V^{(\mathrm{OME})}(p'\boldsymbol{n}', p\boldsymbol{n}) = c\,\mathcal{O}(\boldsymbol{q})\,[D_{\beta_1} + D_{\beta_2}].$$

With the incoming direction along z,

$$\begin{aligned} \boldsymbol{q} &= p\hat{\boldsymbol{e}}_z - p'\boldsymbol{n}' = (-p'\sqrt{1-x^2}\cos\varphi,\ -p'\sqrt{1-x^2}\sin\varphi,\ p - p'x), \\ q^2 &= p^2 + p'^2 - 2pp'x, \qquad \boldsymbol{n}\cdot\boldsymbol{q} = p - p'x, \qquad \boldsymbol{n}'\cdot\boldsymbol{q} = px - p'. \end{aligned} \tag{20}$$

The TOPT functions are

$$D_\beta(p', p, x; m_\phi) = \frac{1}{2\omega_\phi(x)[\omega_\phi(x) + \mu_\beta(p', p)]}, \qquad \omega_\phi(x) = \sqrt{p^2 + p'^2 + m_\phi^2 - 2pp'x}.$$

The μ_β depend on p, p' and the total energy, not on directions [P: Eqs. (76),(77)]. Thus the denominators have no φ dependence, but they *do* retain their x dependence and must stay inside the x integral.

Why the code still performs an azimuthal average. The complete numerator trace is already a scalar. However, its individual Cartesian terms contain powers of $\cos\varphi$, $\sin\varphi$ and $\sqrt{1-x^2}$. The code simplifies their sum by applying the exact average

$$T_{\alpha'\alpha}(x; p', p) \equiv \int_0^{2\pi} \frac{d\varphi}{2\pi}\,\mathrm{Tr}[P^\dagger(\alpha'; \boldsymbol{n}')\,\mathcal{O}(\boldsymbol{q})\,P(\alpha; \hat{\boldsymbol{e}}_z)]. \tag{21}$$

This is an algebraic convenience, not an additional approximation. For example, $\langle\cos^2\varphi\rangle = \langle\sin^2\varphi\rangle = 1/2$ and $\langle\sin\varphi\cos\varphi\rangle = 0$. More generally,

$$\left\langle \cos^{2a}\varphi\,\sin^{2b}\varphi \right\rangle = \frac{(2a-1)!!(2b-1)!!}{(2a+2b)!!},$$

with $(-1)!! = 1$; an odd power of either sine or cosine averages to zero. After the average, every surviving even power of $\sqrt{1-x^2}$ becomes a power of $1-x^2$. Equivalently, one may reduce the scalar directly to the dot products in Eq. (20). Either route gives a polynomial

$$T_{\alpha'\alpha}(x; p', p) = \sum_{m=0}^{m_{\max}} T^{(m)}_{\alpha'\alpha}(p', p)\,x^m.$$

Putting the factors back in their original order gives [D]

$$\begin{aligned} V^{(\mathrm{OME})}_{\alpha'\alpha}(p', p; m_\phi) &= \underbrace{\frac{1}{5}}_{\text{average over } M_J} \underbrace{\frac{1}{2}}_{d\Omega'/4\pi\ \to\ dx/2} c\int_{-1}^{1} dx\,T_{\alpha'\alpha}(x; p', p)\,[D_{\beta_1} + D_{\beta_2}] \\ &= \frac{c}{10}\sum_m T^{(m)}_{\alpha'\alpha}(p', p)\,\Delta^{(\beta_1\beta_2)}_m(p', p; m_\phi), \end{aligned} \tag{22}$$

where, exactly as in the paper's Eq. (D11),

$$\Delta^{(\beta_1\beta_2)}_m(p', p; m_\phi) = \int_{-1}^{1} dx\,x^m\left[D_{\beta_1}(p', p, x; m_\phi) + D_{\beta_2}(p', p, x; m_\phi)\right]. \tag{23}$$

Thus Δ_m is just a name for a one-dimensional integral, not a further projection. Its m is the power of x, *not* a magnetic quantum number or the orbital angular momentum. It contains neither the factor $1/2$ nor $1/5$; both are already in $cT^{(m)}/10$.

Two new denominator combinations appear, $\Delta^{(33)}_m$ and $\Delta^{(44)}_m$, with $\Delta^{(44)}_m(p', p) = \Delta^{(33)}_m(p, p')$. Moments up to $m = 4$ occur [N]. Naively the ${}^5G_2 \to {}^5G_2$ element could contain x^5 ($P_4(x)$ times the x in q^2), but that term cancels, as x^3 cancels in the 1^{++} state $(2,1)J_\ell{=}1$ (Sect. 8) [N].

5.4 A worked example: ${}^5S_2 \to {}^5S_2$

V_{11}: separating the spin trace from the angle integral [D]

For two ${}^5S_2$ projectors, $L = L' = 0$ and $S = J = 2$. The projectors contain no $\boldsymbol{n}$ or $\boldsymbol{n}'$, but the potential still depends on their relative angle through $\boldsymbol{q}$ and the denominators. Write $(S_k)_{ba} = -i\varepsilon_{kba}$ for the spin-1 matrices. Then

$$\boldsymbol{A}\cdot\boldsymbol{q} = \frac{1}{\sqrt{2}}\,\langle\boldsymbol{\epsilon}'|\boldsymbol{S}\cdot\boldsymbol{q}|\boldsymbol{\epsilon}\rangle\,, \qquad (\boldsymbol{A}_1\cdot\boldsymbol{q})(\boldsymbol{A}_2\cdot\boldsymbol{q}) = \tfrac{1}{2}(\boldsymbol{S}_1\cdot\boldsymbol{q})(\boldsymbol{S}_2\cdot\boldsymbol{q})$$

as an operator on the two-vector spin space. The $1/5$ trace is the average over the five $S = 2$ states. Rotational symmetry of this average implies

$$\frac{1}{5}\mathrm{Tr}_{S=2}(S_{1i}S_{2j}) = a\,\delta_{ij}.$$

Contracting $i = j$ determines a without any explicit polarisation vectors:

$$3a = \frac{1}{5}\mathrm{Tr}_{S=2}(\boldsymbol{S}_1\cdot\boldsymbol{S}_2) = 1, \qquad \boldsymbol{S}_1\cdot\boldsymbol{S}_2 = \tfrac{1}{2}[S(S+1) - s_1(s_1+1) - s_2(s_2+1)] = \tfrac{1}{2}(6-2-2) = 1.$$

Consequently,

$$\frac{1}{5}\mathrm{Tr}[P_1^\dagger \mathcal{O} P_1] = \tfrac{1}{2}q_iq_j\,\frac{\delta_{ij}}{3} = \frac{q^2}{6}, \qquad T_{11}(x) = \frac{5}{6}q^2.$$

Notice the distinction: T_{11} is the *unaveraged* trace of Eq. (21), whereas $q^2/6$ already includes $1/5$.
Now insert $c = -g_b^2/(2F_\pi^2)$ and $D_{\beta_1} + D_{\beta_2} = 2D_2$ from Eq. (75):

$$\begin{aligned} V_{11} &= -\frac{g_b^2}{2F_\pi^2}\,\frac{1}{2}\int_{-1}^{1} dx\,\frac{q^2}{6}\,[2D_2] \\ &= -\frac{g_b^2}{24F_\pi^2}\int_{-1}^{1} dx\,\left[(p^2+p'^2) - 2pp'x\right][2D_2] \\ &= \frac{g_b^2}{24F_\pi^2}\left[2pp'\Delta_1^{(22)} - (p^2+p'^2)\Delta_0^{(22)}\right]. \end{aligned}$$

An S-wave projection is therefore an *angular average*, not evaluation at $x = 0$ or $x = 1$. Although both external states are S waves, the x in q^2 produces a Δ_1 moment.
For 1^{++}, $(\boldsymbol{\epsilon}\cdot\boldsymbol{q})(\boldsymbol{\epsilon}'^*\cdot\boldsymbol{q})$ averaged over the three ${}^3S_1$ states gives $q^2/3$; combined with $c = -g_b^2/(4F_\pi^2)$, Eq. (72) gives the same overall coefficient and Eq. (D2), with $\Delta_m^{(12)}$. The S-wave OME of the 2^{++} spin partner differs from the X_b channel *only* through the TOPT denominators: $2D_2$ instead of $D_1 + D_2$.

5.5 A worked example with orbital structure: ${}^1D_2 \to {}^5S_2$

The previous example did not contain direction-dependent projectors. To see where a D wave enters, take V_{16}: incoming $B\bar{B}({}^1D_2)$, outgoing $B^*\bar{B}^*({}^5S_2)$. The new potential, Eq. (4), has

$$c = \frac{g_b^2}{4F_\pi^2}, \qquad \mathcal{O} = (\boldsymbol{\epsilon}_1'^*\cdot\boldsymbol{q})(\boldsymbol{\epsilon}_2'^*\cdot\boldsymbol{q}), \qquad D_{\beta_1} + D_{\beta_2} = 2D_3.$$

Use Cartesian labels a, b for the outgoing vector spins. The outgoing S-wave projector is $P^{(1)}_{ij;ab} = \Pi_{ij,ab}$ of Eq. (15), while the incoming D-wave projector is $P^{(6)}_{ij} = -\sqrt{15/2}\,(n_in_j - \delta_{ij}/3)$. Summing

the outgoing polarisations first gives

$$\sum_{a,b} \Pi_{ij,ab}\, q_a q_b = q_i q_j - \tfrac{1}{3}\delta_{ij} q^2 .$$

The remaining Cartesian contraction is now elementary [D]:

$$\begin{aligned} T_{16} &= -\sqrt{\tfrac{15}{2}} \sum_{i,j} \left(n_i n_j - \tfrac{1}{3}\delta_{ij}\right)\left(q_i q_j - \tfrac{1}{3}\delta_{ij} q^2\right) \\ &= -\sqrt{\tfrac{15}{2}}\left[(\boldsymbol{n}\cdot\boldsymbol{q})^2 - \tfrac{1}{3}q^2\right] \\ &= -\frac{1}{3}\sqrt{\tfrac{15}{2}}\left[3(p - p'x)^2 - (p^2 + p'^2 - 2pp'x)\right] \\ &= -\frac{1}{3}\sqrt{\tfrac{15}{2}}\left[3p'^2x^2 - 4pp'x + (2p^2 - p'^2)\right]. \end{aligned} \tag{24}$$

This expression shows explicitly why choosing $\boldsymbol{n} = \hat{\boldsymbol{e}}_z$ does not discard the incoming D wave: its $n_i n_j - \delta_{ij}/3$ tensor produces the combination $(\boldsymbol{n}\cdot\boldsymbol{q})^2 - q^2/3$. It also shows that no azimuth remains after the full contraction.

Multiplying the polynomial coefficients by $c/10$ gives

$$V_{16} = -\frac{\sqrt{3}\, g_b^2}{24\sqrt{10}\, F_\pi^2}\left[3p'^2\Delta_2^{(33)} - 4pp'\Delta_1^{(33)} + (2p^2 - p'^2)\Delta_0^{(33)}\right],$$

the entry quoted in Sect. 6. For the reversed process V_{61}, the D-wave tensor belongs to $\boldsymbol{n}'$ instead; $(\boldsymbol{n}'\cdot\boldsymbol{q})^2 = (px - p')^2$ interchanges the roles of p and p', and $2D_3$ becomes $2D_4$.

5.6 If the remaining angle integral is evaluated analytically

The reduction to one angle is already complete in Eq. (22); the Δ_m may simply be evaluated by one-dimensional quadrature. The extra change of variables in App. D is only a way to *perform* that last integral. For clarity, consider one denominator at a time:

$$I_m^{(\beta)} = \int_{-1}^{1} dx\, \frac{x^m}{2\omega_\phi(x)[\omega_\phi(x) + \mu_\beta]}, \qquad \Delta_m^{(\beta_1\beta_2)} = I_m^{(\beta_1)} + I_m^{(\beta_2)}.$$

For $\mathcal{B} = 2pp' > 0$, set $\mathcal{A}_\phi = p^2 + p'^2 + m_\phi^2$ and $u = \omega_\phi(x) = \sqrt{\mathcal{A}_\phi - \mathcal{B}x}$. Then [P: Eqs. (D13)–(D16)] [D]

$$x = \frac{\mathcal{A}_\phi - u^2}{\mathcal{B}}, \qquad dx = -\frac{2u}{\mathcal{B}}\, du, \qquad u_- = \sqrt{(p - p')^2 + m_\phi^2}, \quad u_+ = \sqrt{(p + p')^2 + m_\phi^2}.$$

As x goes from -1 to 1, u goes from u_+ down to u_-. Reversing the limits absorbs the minus sign in dx, while its $2u$ cancels the $2\omega_\phi$ in the denominator:

$$I_m^{(\beta)} = \frac{1}{\mathcal{B}^{m+1}} \int_{u_-}^{u_+} du\, \frac{(\mathcal{A}_\phi - u^2)^m}{u + \mu_\beta}. \tag{25}$$

For integer $m \geq 0$, polynomial division leaves a polynomial plus a multiple of $1/(u + \mu_\beta)$, so the answer consists of powers and a logarithm. In particular,

$$I_0^{(\beta)} = \frac{1}{\mathcal{B}}\big[\operatorname{Log}(u_+ + \mu_\beta) - \operatorname{Log}(u_- + \mu_\beta)\big].$$

The μ_β retain the paper's $-i0$ prescription; above a three-body threshold that prescription must also be retained in the logarithms. This substitution changes neither the spin projection nor its normalisation.

At $\mathcal{B} = 0$, use the original integral rather than the apparent $0/0$ form above. Each D_β is then independent of x, so [P: Eq. (D17)] [D]

$$\Delta_m^{(\beta_1\beta_2)}\Big|_{\mathcal{B}=0} = \begin{cases} \dfrac{1}{m+1}\,\Delta_0^{(\beta_1\beta_2)}\Big|_{\mathcal{B}=0}, & m \text{ even}, \\ 0, & m \text{ odd}. \end{cases}$$

5.7 Validation on 1^{++}

The same code applied to the three 1^{++} channels with the projectors of Eq. (D1) reproduces all nine elements of Eqs. (D2)–(D10) exactly, including the assignment of $\Delta_m^{(12)}$, $\Delta_m^{(24)}$, $\Delta_m^{(23)}$, $\Delta_m^{(22)}$ and the time-reversed elements (D5), (D8), (D9) [N].

6 Result: the projected 2^{++} OME

The 2^{++} OME matrix (the analogue of Eqs. (D2)–(D10)) [N]

Channels as in Eq. (1): $1 = {}^5S_2$, $2 = {}^1D_2^*$, $3 = {}^5D_2$, $4 = {}^5G_2$ (all $B^*\bar{B}^*$), $5 = B\bar{B}^*({}^3D_2)$, $6 = B\bar{B}({}^1D_2)$. All $\Delta_m^{(\beta_1\beta_2)} \equiv \Delta_m^{(\beta_1\beta_2)}(p', p; m_\phi)$, with p' (p) the outgoing (incoming) momentum.

$$V_{11} = \frac{g_b^2}{24F_\pi^2}\Big[2pp'\Delta_1^{(22)} - (p^2+p'^2)\Delta_0^{(22)}\Big], \tag{26}$$

$$V_{12} = \frac{g_b^2}{24\sqrt{10}\,F_\pi^2}\Big[3p'^2\Delta_2^{(22)} - 4pp'\Delta_1^{(22)} + (2p^2-p'^2)\Delta_0^{(22)}\Big], \tag{27}$$

$$V_{13} = -\frac{\sqrt{7}\,g_b^2}{24\sqrt{10}\,F_\pi^2}\Big[3p'^2\Delta_2^{(22)} - 4pp'\Delta_1^{(22)} + (2p^2-p'^2)\Delta_0^{(22)}\Big], \tag{28}$$

$$V_{14} = 0, \tag{29}$$

$$V_{15} = -\frac{g_b^2}{8\sqrt{10}\,F_\pi^2}\Big[3p'^2\Delta_2^{(23)} - 4pp'\Delta_1^{(23)} + (2p^2-p'^2)\Delta_0^{(23)}\Big], \tag{30}$$

$$V_{16} = -\frac{\sqrt{3}\,g_b^2}{24\sqrt{10}\,F_\pi^2}\Big[3p'^2\Delta_2^{(33)} - 4pp'\Delta_1^{(33)} + (2p^2-p'^2)\Delta_0^{(33)}\Big], \tag{31}$$

$$V_{22} = -\frac{g_b^2}{24F_\pi^2}\Big[6pp'\Delta_3^{(22)} - 3(p^2+p'^2)\Delta_2^{(22)} - 2pp'\Delta_1^{(22)} + (p^2+p'^2)\Delta_0^{(22)}\Big], \tag{32}$$

$$V_{23} = \frac{g_b^2}{24\sqrt{7}\,F_\pi^2}\Big[3pp'\Delta_3^{(22)} - 3(p^2+p'^2)\Delta_2^{(22)} + pp'\Delta_1^{(22)} + (p^2+p'^2)\Delta_0^{(22)}\Big], \tag{33}$$

$$V_{24} = \frac{g_b^2}{32\sqrt{35}\,F_\pi^2}\Big[35p'^2\Delta_4^{(22)} - 40pp'\Delta_3^{(22)} + 6(2p^2-5p'^2)\Delta_2^{(22)} + 24pp'\Delta_1^{(22)} - (4p^2-3p'^2)\Delta_0^{(22)}\Big], \tag{34}$$

$$V_{25} = 0, \tag{35}$$

$$V_{26} = -\frac{\sqrt{3}\,g_b^2}{48F_\pi^2}\Big[6pp'\Delta_3^{(33)} - 3(p^2+p'^2)\Delta_2^{(33)} - 2pp'\Delta_1^{(33)} + (p^2+p'^2)\Delta_0^{(33)}\Big], \tag{36}$$

$$V_{33} = \frac{g_b^2}{336F_\pi^2}\Big[51pp'\Delta_3^{(22)} - 30(p^2+p'^2)\Delta_2^{(22)} - 11pp'\Delta_1^{(22)} + 10(p^2+p'^2)\Delta_0^{(22)}\Big], \tag{37}$$

$$V_{34} = -\frac{g_b^2}{112\sqrt{5}\,F_\pi^2}\Big[35p'^2\Delta_4^{(22)} - 40pp'\Delta_3^{(22)} + 6(2p^2-5p'^2)\Delta_2^{(22)} + 24pp'\Delta_1^{(22)} - (4p^2-3p'^2)\Delta_0^{(22)}\Big], \tag{38}$$

$$V_{35} = -\frac{g_b^2}{16\sqrt{7}\,F_\pi^2}\Big[3pp'\Delta_3^{(23)} - 3(p^2+p'^2)\Delta_2^{(23)} + pp'\Delta_1^{(23)} + (p^2+p'^2)\Delta_0^{(23)}\Big], \tag{39}$$

$$V_{36} = -\frac{g_b^2}{8\sqrt{21}\,F_\pi^2}\Big[3pp'\Delta_3^{(33)} - 3(p^2+p'^2)\Delta_2^{(33)} + pp'\Delta_1^{(33)} + (p^2+p'^2)\Delta_0^{(33)}\Big], \tag{40}$$

$$V_{44} = \frac{g_b^2}{448F_\pi^2}\Big[35(p^2+p'^2)\Delta_4^{(22)} - 40pp'\Delta_3^{(22)} - 30(p^2+p'^2)\Delta_2^{(22)} + 24pp'\Delta_1^{(22)} + 3(p^2+p'^2)\Delta_0^{(22)}\Big], \tag{41}$$

$$V_{45} = \frac{g_b^2}{16\sqrt{35}\,F_\pi^2}\Big[35p^2\Delta_4^{(23)} - 40pp'\Delta_3^{(23)} - 6(5p^2-2p'^2)\Delta_2^{(23)} + 24pp'\Delta_1^{(23)} + (3p^2-4p'^2)\Delta_0^{(23)}\Big], \tag{42}$$

$$V_{46} = -\frac{\sqrt{3}\,g_b^2}{32\sqrt{35}\,F_\pi^2}\Big[35p^2\Delta_4^{(33)} - 40pp'\Delta_3^{(33)} - 6(5p^2-2p'^2)\Delta_2^{(33)} + 24pp'\Delta_1^{(33)} + (3p^2-4p'^2)\Delta_0^{(33)}\Big], \tag{43}$$

$$V_{55} = \frac{g_b^2}{16F_\pi^2}\,pp'\Big[\Delta_3^{(12)} - \Delta_1^{(12)}\Big], \tag{44}$$

$$V_{56} = V_{66} = 0.$$

The remaining elements follow from time reversal [N],

$$V_{\alpha\alpha'}(p',p;m_\phi) = V_{\alpha'\alpha}(p,p';m_\phi),$$
$$\text{with}\quad \Delta_m^{(23)}(p,p') = \Delta_m^{(24)}(p',p), \qquad \Delta_m^{(33)}(p,p') = \Delta_m^{(44)}(p',p), \tag{45}$$

as for V_{31} and V_{32} in App. D [P: I. 2137]. For example, $V_{51} = -\frac{g_b^2}{8\sqrt{10}F_\pi^2}\big[3p^2\Delta_2^{(24)} - 4pp'\Delta_1^{(24)} + (2p'^2 - p^2)\Delta_0^{(24)}\big]$. The full isoscalar matrix is Eq. (82) unchanged: $V_{\alpha'\alpha}^{(\mathrm{OME})} = 3V_{\alpha'\alpha}^{(\mathrm{OME})}(m_\pi) + \frac{1}{3}V_{\alpha'\alpha}^{(\mathrm{OME})}(m_\eta)$.

Which TOPT combination enters where. This follows from the table of Sect. 3.4 [D]:

out \ in	1	2	3	4	5	6
1–4 ($B^*\bar{B}^*$)	(22)	(22)	(22)	(22)	(23)	(33)
5 ($B\bar{B}^*$)	(24)	(24)	(24)	(24)	(12)	—
6 ($B\bar{B}$)	(44)	(44)	(44)	(44)	—	—

Besides the zeros forced by the missing $BB\phi$ vertex, two elements vanish identically for any masses [N]:

- $V_{14} = V_{41} = 0$. Both channels are $B^*\bar{B}^*$ and use only $2D_2$; the operator conserves J_ℓ (Sect. 8); ${}^5S_2$ is pure $J_\ell = 1$ and ${}^5G_2$ pure $J_\ell = 3$. The G wave couples to the S wave only through the D-wave channels.
- $V_{25} = V_{52} = 0$. The spin part of every OME is a sum of rank-0 and rank-2 tensors in spin space. Rank 0 conserves S, and rank 2 cannot connect $S = 1$ to $S = 0$ (no triangle $1, 2, 0$); cf. $\sum_b \varepsilon_{kab}q_kq_b = 0$ in Eq. (73) contracted with $\delta_{b_1b_2}$.

The S-wave row has the D-wave bracket of Eqs. (D3),(D4) in every entry, and V_{23}, V_{35}, V_{36} have the bracket of Eq. (D7). V_{55} equals Eq. (D10) with $\Delta^{(22)} \to \Delta^{(12)}$ [N].

Where it enters. The total 2^{++} potential has the form of Eq. (83) with $\alpha = 1$ the ${}^5S_2$ channel, and the LSE is Eq. (86) with the six propagators of Sect. 2.2 [P: Eqs. (83),(86)] [D].

7 Checks

What was checked, and how [N]

1. **The method on 1^{++}.** The code that produced Sect. 6, run with Eq. (D1), reproduces Eqs. (D2)–(D10) symbolically, element by element, including the $\Delta^{(\beta_1\beta_2)}$ labels (`pwa.py`).
2. **Projectors.** $\frac{1}{5}\langle \mathrm{Tr}\, P(\alpha)P^\dagger(\beta)\rangle_\Omega = \delta_{\alpha\beta}$ exactly for the four $B^*\bar{B}^*$ projectors, and $= 1$ for ${}^3D_2$ and $B\bar{B}({}^1D_2)$; the same holds for Eq. (D1) with $\frac{1}{3}$ (`checks.py`).
3. **Time reversal**, Eq. (45), symbolically for all 36 elements (`pwa.py`).
4. **Brute force.** For physical masses, $g_b = 0.56$, $F_\pi = 92.2\,\mathrm{MeV}$, $E = -120\,\mathrm{MeV}$ (below all thresholds), $(p,p') = (0.31, 0.57)$ and $(0.83, 0.22)\,\mathrm{GeV}$, and both m_π and m_η, Eq. (80) is evaluated by direct quadrature over *both* spheres with the full TOPT denominators at every $\boldsymbol{q}$, without the reduction of Sect. 5. The maximum deviation from Sect. 6, relative to the largest element, is 2×10^{-12} for 2^{++} and 7×10^{-13} for 1^{++} (on a 30×36 grid per sphere; 1.5×10^{-8} on 20×24, so the agreement is set by quadrature) (`numcheck.py`).
5. **One-loop leading divergences.** In the static HQSS limit Baru *et al.* split the leading UV divergence of the iterated OPE in ${}^3S_1 \to {}^3S_1$ (1^{++}) and ${}^5S_2 \to {}^5S_2$ (2^{++}) into intermediate

partial waves [R: Baru 2016, Table 1]. The weight of channel α is $|V_{\alpha S}(l,0)|^2/\sum_\beta |V_{\beta S}(l,0)|^2$. From Eqs. (D2)–(D4) and Sect. 6 (`checks.py`):

1^{++}	$^3S_1$	$^3D_1$	$^5D_1$			
here = Baru	1/9	2/9	2/3			
2^{++}	$B\bar{B}(^1D_2)$	$B\bar{B}^*(^3D_2)$	$^5S_2$	$^1D_2^*$	$^5D_2$	$^5G_2$
here = Baru	2/15	2/5	1/9	2/45	14/45	0

All nine numbers agree exactly. The two sums are equal, $l^4/16$ in units of $(2D\,g_b^2/F_\pi^2)^2$, as HQSS requires. This tests the normalisation of every projector, the magnitude of the new potential Eq. (4) (the 2/15), and the coupled-channel bookkeeping.

6. **HQSS block structure** (Sect. 8): exact block diagonalisation in the heavy-quark limit, with the $J_\ell = 1, 2$ blocks equal to those of 1^{++}. This tests the relative *signs*, including that of Eq. (4).

Reference values for an implementation. $V_{\alpha'\alpha}(p' = 0.57\,\text{GeV}, p = 0.31\,\text{GeV}; m_\pi)$ for one pion exchange, *without* the isospin factor 3, at $E = -120\,\text{MeV}$, physical masses, $g_b = 0.56$, $F_\pi = 92.2\,\text{MeV}$, in GeV^{-2}; rows α' (out), columns α (in) [N]:

	1	2	3	4	5	6
1	−2.1664	0.1172	−0.3101	0	−0.3548	−0.2067
2	0.9995	−0.0740	0.1654	0.0070	0	−0.0635
3	−2.6445	0.1654	0.1308	−0.0053	−0.2574	−0.3080
4	0	0.1488	−0.1125	−0.0222	0.3139	−0.2859
5	−3.0838	0	−0.2574	0.0143	−0.2001	0
6	−1.8296	−0.0635	−0.3079	−0.0127	0	0

The first column is large because the incoming S wave carries no centrifugal factor p^L at the small p chosen.

8 The heavy-quark spin symmetry limit

For $m_B = m_{B^*}$ all D_β coincide, and so do all $\Delta_m^{(\beta_1\beta_2)} \equiv \Delta_m$ [P: I. 2153]. For 1^{++} the paper records the consequences $V_{13} = -\sqrt{3}V_{12}$ and $2V_{23} = \sqrt{3}(V_{22} - V_{33})$ [P: I. 2153]. These are the statement that OME acts only on the light degrees of freedom, so it conserves $s_{\bar{b}b}$, j_ℓ and J_ℓ. For 2^{++} the orthogonal matrix U to the basis of Sect. 2.3 was found numerically, from the invariant subspaces generated by the coefficient matrices at random momenta, and then identified exactly [N]:

	$^5S_2$	$^1D_2^*$	$^5D_2$	$^5G_2$	$B\bar{B}^*(^3D_2)$	$B\bar{B}(^1D_2)$
$j_\ell = 0$ ($s_{\bar{b}b} = 0$, $L = 2$)	0	$\frac{\sqrt{3}}{2}$	0	0	0	$\frac{1}{2}$
$(0,1)J_\ell$=1	1	0	0	0	0	0
$(2,1)J_\ell$=1	0	$-\frac{1}{2\sqrt{5}}$	$\sqrt{\frac{7}{20}}$	0	$\frac{3}{2\sqrt{5}}$	$\sqrt{\frac{3}{20}}$
$(2,1)J_\ell$=2	0	$-\frac{1}{2\sqrt{3}}$	$-\frac{\sqrt{7}}{2\sqrt{3}}$	0	$\frac{1}{2\sqrt{3}}$	$\frac{1}{2}$
$(2,1)J_\ell$=3	0	$-\sqrt{\frac{7}{60}}$	$\frac{1}{\sqrt{15}}$	0	$-\sqrt{\frac{7}{15}}$	$\sqrt{\frac{7}{20}}$
$(4,1)J_\ell$=3	0	0	0	1	0	0

$(L, j_\ell)J_\ell$ labels $\boldsymbol{J}_\ell = \boldsymbol{L} + \boldsymbol{j}_\ell$ with $s_{\bar{b}b} = 1$. The sign of the $(2,1)J_\ell$=1 row is fixed so that it matches $|(2,1)J_\ell{=}1\rangle = -\frac{1}{2}\,|^3D_1\rangle + \frac{\sqrt{3}}{2}\,|^5D_1\rangle$ of Eq. (20) through the S-wave coupling; the other row signs are conventions. The first row is the $0_H \otimes 0_\ell$ combination of Eq. (2). In this basis, for all p, p' and

every m separately [N]:

$$U\,V^{(\mathrm{OME})}\,U^{T} = V^{(j_\ell=0)} \oplus V_{2\times2}^{(J_\ell=1)} \oplus V^{(J_\ell=2)} \oplus V_{2\times2}^{(J_\ell=3)} \tag{46}$$

(off-block elements below 10^{-15} of the largest entry). This refines the decomposition $A(2\times2)\oplus D(1\times1)\oplus E(3\times3)$ of Baru *et al.* [R: Baru 2016, Eq. (39); Baru 2017, Eq. (3.14)]: their E splits further into $J_\ell=3$ and $j_\ell=0$.

- $V^{(J_\ell=1)}$ and $V^{(J_\ell=2)}$ are *identical* to the 1^{++} blocks in the basis of Eq. (20) (deviation $<10^{-16}$) [N]. Explicitly, with $|S\rangle\equiv|(0,1)1\rangle$, $|D\rangle\equiv|(2,1)1\rangle$ and $V_{DS}(p',p)=V_{SD}(p,p')$,

$$\begin{aligned}
V_{SS}^{(J_\ell=1)} &= \frac{g_b^2}{24F_\pi^2}\Big[2pp'\Delta_1-(p^2+p'^2)\Delta_0\Big],\\
V_{SD}^{(J_\ell=1)} &= -\frac{g_b^2}{12\sqrt{2}F_\pi^2}\Big[3p'^2\Delta_2-4pp'\Delta_1+(2p^2-p'^2)\Delta_0\Big],\\
V_{DD}^{(J_\ell=1)} &= \frac{g_b^2}{48F_\pi^2}\Big[3(p^2+p'^2)\Delta_2-4pp'\Delta_1-(p^2+p'^2)\Delta_0\Big],\\
V^{(J_\ell=2)} &= \frac{g_b^2}{16F_\pi^2}\Big[4pp'\Delta_3-3(p^2+p'^2)\Delta_2+(p^2+p'^2)\Delta_0\Big].
\end{aligned}$$

  This is the OME version of the statement that the 2^{++} $B^*\bar B^*$ state is the spin partner of the 1^{++} $B\bar B^*$ one [R: Baru 2016, Sect. 3.2].

- $V^{(j_\ell=0)} = -\frac{g_b^2}{16F_\pi^2}\big[6pp'\Delta_3-3(p^2+p'^2)\Delta_2-2pp'\Delta_1+(p^2+p'^2)\Delta_0\big]$. This is exactly $(-3)\times\frac{g_b^2}{24F_\pi^2}\int_{-1}^{1}dx\,P_2(x)\,(-q^2)\,[2D]$, while $V_{11}=\frac{g_b^2}{24F_\pi^2}\int_{-1}^{1}dx\,P_0(x)\,(-q^2)\,[2D]$. So it is the central OME in $L=2$ with $\langle\boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_2\rangle=-3$ for a light-spin singlet, versus $+1$ for the triplet that gives V_{11} [N]. The singlet has no tensor force.

- $V^{(J_\ell=3)}$ with $|D'\rangle\equiv|(2,1)3\rangle$, $|G\rangle\equiv|(4,1)3\rangle={}^5G_2$ and $V_{GD'}(p',p)=V_{D'G}(p,p')$ [N]:

$$\begin{aligned}
V_{D'D'}^{(J_\ell=3)} &= \frac{g_b^2}{112F_\pi^2}\Big[10pp'\Delta_3-3(p^2+p'^2)\Delta_2-6pp'\Delta_1+(p^2+p'^2)\Delta_0\Big], \qquad V_{GG}^{(J_\ell=3)}=V_{44},\\
V_{D'G}^{(J_\ell=3)} &= -\frac{\sqrt{3}\,g_b^2}{112F_\pi^2}\Big[35p'^2\Delta_4-40pp'\Delta_3+6(2p^2-5p'^2)\Delta_2+24pp'\Delta_1-(4p^2-3p'^2)\Delta_0\Big].
\end{aligned}$$

Element-level consequences, the analogues of the two 1^{++} relations [N]. Item 4 and $V_{66}=0$ hold for any masses; the others need $m_B=m_{B^*}$.

1. $V_{66}=0$ and $V_{\alpha6}=-\sqrt{3}\,V_{\alpha2}$ for $\alpha=1,3,4,5$: the $s_{\bar bb}=0$ state decouples from everything with $s_{\bar bb}=1$. Also $V_{26}=\frac{\sqrt3}{2}V_{22}$.
2. $V_{12}:V_{13}:V_{15}:V_{16}=1:-\sqrt7:-3:-\sqrt3$: the S wave couples only to the $L=2$, $J_\ell=1$ combination, the $(2,1)1$ row of U.
3. $V_{11}^{(2^{++})}=V_{11}^{(1^{++})}$.
4. $V_{14}=0$.

Phases of the 2^{++} projectors versus standard recoupling [N]

The standard recoupling

$$\langle(Lj_\ell)J_\ell,s_{\bar bb};J|L,(j_\ell s_{\bar bb})S;J\rangle=(-1)^{L+j_\ell+s_{\bar bb}+J}\sqrt{(2J_\ell+1)(2S+1)}\begin{Bmatrix}L & j_\ell & J_\ell\\ s_{\bar bb} & J & S\end{Bmatrix}$$

reproduces Eq. (20) for 1^{++} (`notes/s_channel`, Sect. 2.7), and the paper's projectors (D1) are consistent with it: Eq. (20) block-diagonalises Eqs. (D2)–(D10) [N]. For 2^{++} the magnitudes

of the $L = 2$ rows of U equal these coefficients, with the $S = 0$ entry taken on the $(1_H \otimes 1_\ell)_0$ combination $\frac{\sqrt{3}}{2}\,|B\bar{B}(^1D_2)\rangle - \frac{1}{2}\,|B^*\bar{B}^*(^1D_2)\rangle$ of Eq. (2). The signs, however, come out as $(-1)^{J_\ell+1}(-1)^S$ times the standard ones. The row factor is a convention. The column factor means that the $^3D_2$ projector (B7) carries the opposite phase, relative to $^5D_2$ (B13) and to the $(1_H \otimes 1_\ell)_0$ part, from the one the standard table assumes. If the D-wave couplings of the χ_{b2} (the analogue of Eqs. (17), (22)) are ever built from standard CG/6j tables, flip the sign of the $^3D_2$ entries to stay consistent with the OME matrix. This is the same kind of adjustment as the $^5D_1$ footnote at l. 436 [P: l. 436]. At LO nothing depends on it, because only $^5S_2$ couples to the bottomonia and to C.

9 What else the 2^{++} LSE needs (outside OME)

Only listed here; none of it is part of the derivation above.

- **Potential and propagators.** Eq. (83) with $\alpha = 1 = {}^5S_2$; the propagators of Sect. 2.2; channels 5 and 6 are open in the energy range of a $B^*\bar{B}^*$-threshold state [D].
- **s-channel coupling.** At LO only the S wave couples. The factors in g [P: Eqs. (14)–(18)] come from the light degrees of freedom: flavour, the $\Sigma_g^{+\prime}$ overlap, and the S-wave share of $\Sigma_g^{+\prime}$. The heavy spin is a spectator. The S-wave channel is the pure state $(1_H \otimes 1_\ell)_J$ both for $J = 1$ ($B\bar{B}^*$, C-even) and for $J = 2$ ($B^*\bar{B}^*$, $S = 2$), Eq. (2), so no extra spin factor appears. (For $J = 0$ the same state would be split over $B\bar{B}$ and $B^*\bar{B}^*$.) [D]. The S-wave share uses the same Step-2 coefficients as Eq. (21), $|\Sigma_g^{+\prime}; J_\ell{=}1\rangle = \frac{1}{\sqrt{3}}\,|(0,1)1\rangle - \sqrt{\frac{2}{3}}\,|(2,1)1\rangle$. With the rows of U this gives the 2^{++} analogue of Eq. (17) [N]:

$$|[\bar{b}b]_1(nP); J{=}2\rangle \to \tfrac{1}{\sqrt{3}}\,|B^*\bar{B}^*(^5S_2)\rangle + \tfrac{1}{\sqrt{30}}\,|B^*\bar{B}^*(^1D_2)\rangle - \sqrt{\tfrac{7}{30}}\,|B^*\bar{B}^*(^5D_2)\rangle - \sqrt{\tfrac{3}{10}}\,|B\bar{B}^*(^3D_2)\rangle - \tfrac{1}{\sqrt{10}}\,|B\bar{B}(^1D_2)\rangle\,.$$

  The same code reproduces Eq. (17) for 1^{++} [N]. The S-wave share is $1/\sqrt{3}$ as in Eq. (17), so $g = 57\,\text{MeV}$ carries over [D]. The phases are those of Eqs. (6)–(11).
- **Bare tower.** The $\chi_{bJ}(nP)$ masses are the eigenvalue $M^{(\bar{b}b)}$ of the spin-independent Hamiltonian plus $\Delta M_{\text{SD}}[\chi_{bJ}]$ [P: l. 1977, Eq. (B4)]. So at LO the $\chi_{b2}(nP)$ share R_{1n} with the $\chi_{b1}(nP)$, while the shift $\Delta M_{\text{SD}}[\chi_{b2}] = \langle V_{\text{SL}}\rangle - \frac{1}{5}\langle V_{\text{ST}}\rangle$ differs from that of χ_{b1}. V_0 and C must therefore be refitted to the $\chi_{b2}(nP)$ masses.
- **Widths.** A state near the $B^*\bar{B}^*$ threshold can decay into $B\bar{B}^*$ and $B\bar{B}$ in D waves through V_{15}, V_{16}, V_{35}, …. With nonperturbative pions Baru *et al.* find $\Gamma \approx 50\,\text{MeV}$ for the charm 2^{++} partner and a few MeV for the isovector bottom W_{b2} [R: Baru 2016, abstract and Sect. 3.3; Baru 2017, Table 1].

10 Porting to `HadronDynamics.jl`

The 1^{++} implementation is hard-wired to three channels and four moments. What a 2^{++} version needs, file by file [N]:

- `projection.jl` (in `src/models/BOchiEFT/`, lines 11–29): coefficients of $x^0, \ldots, x^3$ for the three 1^{++} channels. They multiply $-g_b^2/(8F_\pi^2)$ and the summed moments (`exchange.jl`, lines 70–74). For 2^{++} the 6×6 table with $x^0, \ldots, x^4$ is printed by `pwa.py`, section 3, already in this convention (`r` $= p_{\text{out}}$, `p` $= p_{\text{in}}$).

- The code's $B^*\bar{B}^*({}^5D_1)$ basis vector has the opposite sign to Eq. (D1): its $(1,3)$, $(3,1)$, $(2,3)$, $(3,2)$ entries are minus Eqs. (D4), (D8), (D7), (D9), all others agree (`checks.py`, section 5). So $V_{13} = +\sqrt{3}V_{12}$ in the HQSS limit. The comment there says "Condon–Shortley LS phases". This is harmless, but do not compare signs between the code's 1^{++} table and Eq. (D1) or Baru's projectors without accounting for it.
- `exchange.jl`, lines 1–39 (`tc_alphas`): the map from channel pair to $(\mu_{\beta_1}, \mu_{\beta_2})$. Add (μ_3, μ_3) for $B^*\bar{B}^* \leftarrow B\bar{B}$ and (μ_4, μ_4) for $B\bar{B} \leftarrow B^*\bar{B}^*$; $B\bar{B} \leftarrow B\bar{B}$ and $B\bar{B} \leftrightarrow B\bar{B}^*$ have no exchange.
- `ExchangeMoments.jl` (in `src/common/`), line 3: `SD_MOMENT_ORDER = 3` must become 4. `_sd_one_moment_bzero` (lines 9–13) hard-codes four entries (the generalisation is $2/(m+1)$ times the base value for even m, 0 for odd m), and the quadrature branch of `tc_moments` (`exchange.jl`, lines 44–55) allocates four.
- `BOchiEFT.jl`, lines 135–139: `channels` is limited to 1–3, and the three `ContinuumChannels` are 1^{++} specific. 2^{++} needs six, with thresholds $+\Delta m$, 0, $-\Delta m$ and reduced masses $m_{B^*}/2$, μ, $m_B/2$. The switches `tensor_scale` and `third_scale` of `ome_potential` (`exchange.jl`, lines 76–94) are defined through the 1^{++} channel numbers.

11 Caveats and open points

1. **Phases.** Three conventions are in play: Baru's Cartesian projectors (used here, consistent with Eq. (D1)), the standard $6j$ recoupling (consistent with Eq. (D1) for 1^{++}, but not for ${}^3D_2$ in 2^{++}; Sect. 8), and the code's 1^{++} table (${}^5D_1$ flipped). None affects observables, but they matter whenever matrices from different sources are combined.
2. **Sign of Eq. (4).** It is tied to $\mathcal{C}V\mathcal{C}^{-1} = -\bar{V}$. In the convention of Baru *et al.* it has the opposite sign relative to Eqs. (72)–(75); that is equivalent to a phase of the $B\bar{B}$ basis state (Sect. 3.3).
3. **Contact terms.** The paper argues that no S–D counterterm is needed at LO because, unlike in the Z_b system, "there are no lower-lying channels" [P: l. 1075–1080]. In 2^{++} there are: $B\bar{B}^*({}^3D_2)$ and $B\bar{B}({}^1D_2)$ lie 46 and 92 MeV below the S-wave threshold and are reached by the tensor force. The cutoff independence obtained with a single C must be rechecked for 2^{++} before that argument is reused.
4. **The G wave.** In the heavy-quark limit ${}^5G_2$ lives only in the $J_\ell = 3$ block and cannot affect the $J_\ell = 1$ partner of the X_b [R: Baru 2016, footnote 4]. With $\Delta m \neq 0$ it couples to the S wave through the D-wave channels (with different Δ's), so dropping it is exact only in that limit. It costs nothing to keep.
5. **Numerics of Δ_4.** The analytic moments are $\propto \mathcal{B}^{-(m+1)}$ times cancelling combinations. The code already switches to quadrature or extended precision at small $\mathcal{B}$, with guard bits that scale with `SD_MOMENT_ORDER`; this should be retested at $m = 4$.
6. **Not derived here:** the static/local r-space form of the 2^{++} OME (the analogue of Eqs. (25)–(27), which would now include $V_{\Sigma_u^-}$), the s-channel and contact pieces, and any numerical result for χ_{b2} states.

12 Self-check questions

Work these out before reading the answers in Appendix C.

1. Why is there no OME in $B\bar{B} \to B\bar{B}$ and in $B\bar{B} \leftrightarrow B\bar{B}^*$, but there is in $B\bar{B} \to B^*\bar{B}^*$?
2. Why do both time orderings of $B\bar{B} \to B^*\bar{B}^*$ come with μ_3?

3. Why is $B\bar{B}^*(^3S_1)$ absent from 2^{++}, and why is there no $B\bar{B}^*(^3G_2)$?
4. Why does V_{25} vanish for any meson masses, while V_{15} does not?
5. Why is $V_{14} = 0$ exact even with $m_B \neq m_{B^*}$, whereas the $B\bar{B}(^1D_2)$ decoupling of Sect. 8 holds only in the heavy-quark limit?
6. Show that $\frac{1}{2}\,|B\bar{B}(^1D_2)\rangle + \frac{\sqrt{3}}{2}\,|B^*\bar{B}^*(^1D_2)\rangle$ decouples in the heavy-quark limit, and explain the factor -3 in its OME.
7. Why are moments up to $m = 4$ enough, although a G wave has $P_4(x)$ and $\boldsymbol{qq}$ adds one power of x?
8. Which pieces of the 2^{++} OME are the same as in 1^{++} when $m_B = m_{B^*}$, and where does HQSS breaking enter?
9. What changes in Sect. 6 with Baru's charge-conjugation convention? Can any observable change?
10. Why does the coupling $g = 57\,\text{MeV}$ carry over from χ_{b1} to χ_{b2} at LO?

A Notation and numbers

symbol	meaning	value / source
M_{th}, E	$m_B + m_{B^*}$; $E = M - M_{\text{th}}$	10.604 GeV [P: Eqs. (3),(4)]
Δm	$m_{B^*} - m_B$	46 MeV [P: Eq. (3)]
$\alpha = 1,\ldots,6$	${}^5S_2$, ${}^1D_2^*$, ${}^5D_2$, ${}^5G_2$ ($B^*\bar{B}^*$), ${}^3D_2$ ($B\bar{B}^*$), ${}^1D_2$ ($B\bar{B}$)	Eq. (1)
D_β, μ_β	TOPT denominators, $\beta = 1$: $B\bar{B}\phi$, 2: $B^*\bar{B}^*\phi$, 3/4: $B\bar{B}^*\phi$	[P: Eqs. (76),(77)]
$\Delta_m^{(\beta_1\beta_2)}$	$\int_{-1}^{1} x^m [D_{\beta_1} + D_{\beta_2}]dx$	[P: Eq. (D11)]; Eq. (23)
$\Delta_m^{(33)}$, $\Delta_m^{(44)}$	combinations of the new transition; $\Delta_m^{(44)}(p',p) = \Delta_m^{(33)}(p,p')$	Sect. 5
v_{ij}, v_{ijkl}, S_{ij}	$L = 2$, $L = 4$ tensors; $S = 2$ tensor	[R: Baru 2019, Eqs. (B15)–(B18)]
$s_{\bar{b}b}$, j_ℓ, J_ℓ	heavy-pair spin, light spin, $\boldsymbol{L} + \boldsymbol{j}_\ell$	[P: l. 489–495]
g_b, F_π	axial coupling, pion decay constant	0.56, 92.2 MeV [P: Eq. (66), l. 984]
m_π, m_η	exchanged masses	138, 548 MeV [P: Eq. (6)]

B Reproduction scripts

The [N] results come from the Python scripts in `code/` (SymPy/NumPy). Run from inside `code/` with `uv run --no-project --with numpy --with sympy python <script>`:

- `ome.py`: the library. It contains the projectors (Eq. (D1) and Baru 2019 App. B), the six spin structures with prefactors and TOPT combinations (the table of Sect. 3.4), and the symbolic projection at fixed $\boldsymbol{n} = \hat{\boldsymbol{e}}_z$ with an exact φ' average;
- `pwa.py` (a few seconds): 1^{++} versus Eqs. (D2)–(D10); time reversal; the 2^{++} elements in LaTeX (the source of Sect. 6); the same as coefficient tuples in the convention of `projection.jl`;
- `checks.py` (a few seconds): projector normalisation and orthogonality; the Baru 2016 Table 1 weights; the heavy-quark-limit blocks, the matrix U, the comparison with 1^{++} and with the $6j$ coefficients, the $j_\ell = 0$ identity, the $\Sigma_g^{+\prime}$ content; the identically vanishing elements; a transcription of `projection.jl` compared with Eq. (D1) (Sect. 10); the momenta of Sect. 2.2;
- `numcheck.py` (about 30 s; 3 min with `NUMCHECK_GRID=30,36`): the brute-force double-sphere quadrature with full TOPT denominators, and the reference values of Sect. 7.

The scripts were not run against `HadronDynamics.jl` itself (`notes/s_channel`, App. B, explains why); the comparison with `projection.jl` in Sect. 10 uses a transcription of its coefficients (`checks.py`, section 5).

C Answers to the self-check questions

1. One-meson exchange needs a vertex on each line. $B \to B\phi$ is forbidden by parity ($0^- \to 0^-0^-$ with $L = 0$). $B\bar{B} \to B\bar{B}^*$ would need one on the B line; $B\bar{B} \to B^*\bar{B}^*$ has $B \to B^*$ on both lines.
2. In each ordering the intermediate state has the pseudoscalar that has not yet emitted or absorbed (incoming momentum p) and the vector that already has (outgoing p'): $E_B(p) + E_{B^*}(p') - M = \mu_3$.
3. ${}^3S_1$ has $J = 1$. $B\bar{B}^*$ with $S = 1$ and $L = 4$ has $J \geq 3$.
4. The spin part of OME is rank $0 \oplus$ rank 2 in spin space; neither connects $S = 1$ to $S = 0$. V_{15} connects $S = 1$ to $S = 2$, which rank 2 allows.
5. V_{14} involves only $B^*\bar{B}^*$ states and only $2D_2$; no Δ of a different channel can spoil the J_ℓ selection rule. The $B\bar{B}$ decoupling needs $V_{\alpha 6} = -\sqrt{3}V_{\alpha 2}$, which compares $\Delta^{(33)}$ with $\Delta^{(22)}$ (or $\Delta^{(44)}$ with $\Delta^{(22)}$); these differ once $m_B \neq m_{B^*}$.
6. By Eq. (2) it is $0_H \otimes 0_\ell$. OME conserves $s_{\bar{b}b}$, so it cannot mix with the $s_{\bar{b}b} = 1$ states. In the light-spin singlet $(\boldsymbol{\sigma}_1 \cdot \boldsymbol{q})(\boldsymbol{\sigma}_2 \cdot \boldsymbol{q}) \to -q^2$, versus $+\frac{1}{3}q^2$ (central part) in the triplet, hence -3 relative to the triplet central force.
7. For $L = J_\ell + 1$ the x^{L+1} terms cancel between the central and tensor parts; the $J_\ell = 3$, $L = 4$ state is of this type, as is $(2, 1)1$ in 1^{++}, whose x^3 term cancels. The other G-wave elements are D–G, with at most x^4.
8. The $J_\ell = 1$ and $J_\ell = 2$ blocks, in particular V_{11}. HQSS breaking enters only through the D_β: $2D_2$ instead of $D_1 + D_2$ for the S wave, $\Delta^{(23)}, \Delta^{(24)}, \Delta^{(33)}, \Delta^{(44)}$ for the channel couplings, and the propagators G_α.
9. $V_{\alpha 6}$ and $V_{6\alpha}$ change sign; all else stays. That is a change of phase of the $B\bar{B}$ basis vector, so no observable changes.
10. The mixing acts on the light degrees of freedom and leaves the heavy spin alone. Flavour factor, $\Sigma_g^{+\prime}$ overlap and the $1/\sqrt{3}$ S-wave share of $\Sigma_g^{+\prime}$ ($J_\ell = 1$) are J independent.

References

[1] Y. Zhang, F.-K. Guo, C. Hanhart, A. Nefediev, P. Pütz, *Born–Oppenheimer chiral effective field theory and its application to X_b* (this project, `Xb.tex`).

[2] V. Baru, E. Epelbaum, A. A. Filin, C. Hanhart, U.-G. Meißner, A. V. Nefediev, *Heavy-quark spin symmetry partners of the $X(3872)$ revisited*, PLB 763 (2016) 20, arXiv:1605.09649.

[3] V. Baru, E. Epelbaum, A. A. Filin, C. Hanhart, A. V. Nefediev, *Spin partners of the $Z_b(10610)$ and $Z_b(10650)$ revisited*, JHEP 06 (2017) 158, arXiv:1704.07332.

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[5] A. E. Bondar *et al.*, *Heavy quark spin structure in Z_b resonances*, PRD 84 (2011) 054010, arXiv:1105.4473.

[6] `notes/s_channel/s_channel_note.tex` (this project): quantum-number guide, Sect. 2.

[7] `HadronDynamics.jl` (this project), `src/models/BOchiEFT/{projection,exchange,BOchiEFT}.jl`, `src/common/ExchangeMoments.jl`.